Software Test Plan (STP) - Chat Application

Project: Real-Time Chat Application (Socket Programming)

Version: 1.1

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Status: Draft

Revision History

Version	Date	Author	Change Summary
1.0	04-10-2026	Team 5	Initial test plan with test cases and traceability to SRS v1.0
1.1	07-10-2026	Team 5	Added team member names, SRNs and roles

1. Introduction

Purpose: This document defines the test plan for the Real-Time Chat Application v1.0. It states the objectives, scope, strategy, environment, schedule, responsibilities and the test cases that will be used to verify the functional and non-functional requirements of the SRS.

Scope: Testing covers registration and authentication, one-to-one messaging, group chat, presence and typing, chat history, delivery status and unread notifications, profile management and user search, plus the performance (PERF), safety (SAF) and security (SEC) requirements, the business rules (BR) and the quality attributes of usability and reliability. Third-party library internals and the hosting infrastructure are excluded (Section 4).

References: SRS v1.0 (Team 5, 04-Oct-2026) including Appendix C; Software Architecture and Design Specification (SAD) v1.0; IEEE Std 829 (Software Test Documentation); OWASP Top 10 and OWASP Web Security Testing Guide; RFC 6455.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design), RTM (Requirements Traceability Matrix), TC (Test Case), JWT (JSON Web Token), WS/WSS (WebSocket / over TLS), UAT (User Acceptance Test), XSS (Cross-Site Scripting).

2. Test Items

- Auth Module (registration, login, lockout, JWT, WebSocket authentication, logout)
- Messaging Module (one-to-one send, receive, offline queue, validation, sanitising)
- Group Module (create, membership, admin actions, fan-out, authorisation)
- Presence Module (online/offline, typing, heartbeat) and WebSocket client reconnection
- History Module (persistence, paging, missed-message re-sync)
- Status/Notification Module (SENT/DELIVERED/READ, unread counters, notifications)
- Profile/Search Module (profile edit, change password, user search)
- REST API, WebSocket Gateway, Data Access Layer and the web client UI

3. Features to be Tested

Features are mapped to SRS requirement identifiers. Each is covered by the test cases listed in Appendix A and in the RTM (Section 13).

Feature (SRS section)	Requirement IDs	Type
5.1 User Registration and Authentication	REQ-1 to REQ-7	Functional
5.2 One-to-One Real-Time Messaging	REQ-8 to REQ-13	Functional
5.3 Group Chat	REQ-14 to REQ-18	Functional
5.4 Presence, Typing and Connection Handling	REQ-19 to REQ-22	Functional
5.5 Chat History and Persistence	REQ-23, REQ-24	Functional
5.6 Delivery Status and Unread Notifications	REQ-25 to REQ-28	Functional
5.7 Profile Management and User Search	REQ-29 to REQ-31	Functional
6.1 Performance Requirements	PERF-1 (latency), PERF-2 (100 connections), PERF-3 (login time), PERF-4 (history load), PERF-5 (page load)	Non-functional
6.2 Safety Requirements	SAF-1 (write before ack), SAF-2 (rate limit), SAF-3 (graceful failure), SAF-4 (backup/restore)	Non-functional
6.3 Security Requirements	SEC-1 to SEC-8 (see Section 5.1)	Non-functional
6.4 Quality attributes	Usability (first message in 2 minutes, 90% of users), Reliability (4-hour run, no message loss)	Non-functional
6.5 Business Rules	BR-1 to BR-6	Functional

4. Features Not to be Tested

- Internals of third-party libraries and services (Node.js, Express, ws, Mongoose, bcrypt algorithm, MongoDB engine); only their correct use is tested.
- Browser WebSocket implementations and browsers older than the latest two versions of Chrome, Firefox, Edge and Safari.
- Certificate issuance by Let's Encrypt or the hosting provider's infrastructure; only the resulting TLS configuration is tested.
- Features declared out of scope in SRS 1.3: voice/video calls, file and media sharing, end-to-end encryption, native mobile apps, message editing, external messaging integrations.
- Horizontal scaling and clustering (single-server deployment in v1.0); load above 100 simultaneous connections.
- SAF-5: external safety certification (the application is not safety-critical).

5. Test Approach / Strategy

Levels:

Level	What is tested	Tools	Owner
Unit	Functions of each module in isolation (validation, hashing, token handling, status transitions, rate limiter); target 70% coverage of server logic	Jest	Each member, own modules
Integration	REST endpoints and WebSocket flows with a real test MongoDB; module interaction (messaging with status, group with messaging)	Jest + Supertest, ws test clients, Postman/Newman	Module owners, cross-reviewed
System	End-to-end flows through the browser UI with several clients; non-functional checks (performance, reliability, usability)	Selenium WebDriver, k6/Artillery, browser DevTools	All members

Level	What is tested	Tools	Owner
Acceptance (UAT)	Demonstration of the SRS use cases to the project guide and sample users	Manual	Team with guide

Types:

- Functional testing of every REQ (positive, negative and boundary cases, for example 2,000 versus 2,001 characters).
- Regression testing: the automated unit and integration suites run on every pull request and before each demonstration.
- Performance and load testing: latency distribution, 100 simulated clients, login time, history load, page load.
- Security testing: see Section 5.1.
- Reliability testing: heartbeat, reconnection, server crash, database outage, 4-hour soak run.
- Usability and compatibility testing: first-time-user task, 360 px layout, Chrome, Firefox, Edge.

Entry criteria: SRS and SAD baselined; a stable build of the module is merged to the main branch; test database and test users available; test environment set up as in Section 6.

Exit criteria: 100% of planned test cases executed; all test cases for High-priority requirements (REQ-1 to REQ-18, REQ-23, REQ-24) passed; no critical or high-severity defects open; PERF targets met; unit coverage of server logic at least 70%; acceptance criteria demonstrated to the guide.

5.1 Security Validation

Security validation verifies the security objectives and requirements SEC-1 to SEC-8 and the threats of the STRIDE model in the SAD (section 3.9).

Check	Method / tool	Requirement	Test case
Authentication: wrong, missing, tampered, expired tokens (including alg "none" and modified payload) on REST and WebSocket	Postman, script with ws client, jwt tampering script	SEC-1, SEC-7, REQ-6	TC-AUTH-03, TC-AUTH-05
Password storage: only salted bcrypt, cost 10 or higher; no hashes, tokens or passwords in logs or responses	Database inspection, log search	SEC-2, REQ-3	TC-AUTH-01, TC-SEC-01
Brute-force protection: 5 failures lock the account for 15 minutes; generic error message	Script / Postman runner	SEC-7, REQ-5, BR-6	TC-AUTH-04
Transport security: TLS 1.2+ only; no plain http/ws in deployment	testssl.sh, openssl s_client	SEC-3	TC-SEC-02
Input validation and injection: XSS payload set, NoSQL operator injection, malformed JSON and oversized frames (fuzzing of WebSocket events)	Payload lists, script-based fuzzer, OWASP ZAP baseline scan	SEC-4, REQ-13, SAF-2	TC-MSG-05, TC-SEC-04
Authorisation (IDOR): accessing other users' conversations by ID over REST and WebSocket; non-admin group actions	Postman with two accounts, ws script	SEC-5, BR-3, BR-4, REQ-18	TC-GRP-03, TC-GRP-04
Cross-site WebSocket hijacking: handshake from a foreign Origin	ws script with forged Origin header	SEC-6	TC-SEC-03
Denial-of-service resistance: message rate limit, frame size limit, dead-connection clean-up	k6/Artillery, ws script	SAF-2, REQ-21	TC-SEC-04, TC-PRS-03
Privacy and OWASP Top 10 review; dependency vulnerabilities; secrets not in the repository	Checklist review, npm audit, git secret scan	SEC-8	TC-SEC-05

6. Test Environment

Hardware: Team laptops/PCs; one server machine with 2 GB RAM or more for the load test; an Android or iOS phone (or browser emulation) for 360 px testing.

Software: Node.js 20 LTS, MongoDB 6.x (separate test database chatapp_test), application v1.0 from the Git repository; browsers Chrome, Firefox and Edge (latest two versions); Windows 10/11 and Ubuntu 22.04+.

Tools: Jest and Supertest (unit/integration), Selenium WebDriver (UI automation), Postman/Newman (REST), k6 or Artillery (WebSocket load), OWASP ZAP and testssl.sh (security), npm audit, ESLint, Git with GitHub Actions or manual npm test (regression), GitHub Issues (defect tracking).

Test data: Scripted seed of dummy users (alice, bob, carol, dave, ...), private conversations and groups, a conversation with 120 messages for paging tests, and 100 generated load-test accounts. No real personal data is used.

7. Test Schedule

Dates after the Part-1 deadline are proposed and will be aligned with the sprint calendar and the guide's review dates.

Milestone	Date / timing
Test plan and test case design (Part-1 deliverable)	07-Oct-2026
Test environment and seed data set up	Sprint 1
Unit tests written together with code of each module	Every sprint (continuous)
Integration testing after each module is merged	Sprints 1 to 3
System testing, security validation, load and soak tests	After feature freeze (end of sprint 3)
Regression run, defect closure and UAT / demo rehearsal	Final sprint, before the demonstration
Test summary report	With final submission

8. Test Deliverables

- Test Plan (this document) and the test cases in Appendix A
- Automated test scripts (Jest, Supertest, Selenium, k6/Artillery) in the repository /tests folder
- Test data seed scripts
- Test execution logs and coverage reports
- Defect reports (GitHub Issues)
- Updated RTM with test case results
- Test Summary Report

9. Roles and Responsibilities

Every member designs and executes tests for the modules they own and reviews another member's tests, following the ownership in the SRS Team Profile.

Role	Name	Responsibility
Test Lead (rotating per sprint)	Rotates among the four team members	Maintains this plan and the RTM, coordinates execution, prepares the summary report
Security tester / Auth owner	Gagan Eswar (PES1UG24CS165)	Tests for REQ-1 to REQ-7, SEC-1 to SEC-8, security validation (5.1)
Messaging and Performance tester	Chetha Samrutha (PES1UG24CS131)	Tests for REQ-8 to REQ-13, REQ-23, REQ-24, PERF-1 to PERF-5, load tests

Role	Name	Responsibility
Group and Usability tester	Harsh Kalekar (PES1UG24CS183)	Tests for REQ-14 to REQ-18, REQ-29 to REQ-31, usability and compatibility tests
Reliability tester	Dhanush K S (PES1UG24CS148)	Tests for REQ-19 to REQ-22, REQ-25 to REQ-28, SAF-1 to SAF-4, soak test
Project guide		Reviews the plan, attends UAT, approves readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in a stable build of a module	Agree API contracts (SAD 4.3) early; test against mocks; merge small increments each sprint
WebSocket timing makes tests flaky	Use explicit waits and event-based assertions, not fixed sleeps; run critical tests several times
Load test machine or network limits distort results	Run the server and the load generator on separate machines; document the setup with the results
Test environment or database corrupted	Seed scripts recreate data in minutes; backup before each demo
Limited time and team size	Prioritise High-priority requirements and security tests first; automate regression
Security tests miss real-world issues	Use OWASP checklist plus ZAP scan; cross-review by a different team member

11. Assumptions & Dependencies

- The SRS and SAD are stable; changes are communicated and the RTM is updated.
- A MongoDB instance (local or Atlas free tier) is available for the test database.
- Configuration allows a shortened lock time and token expiry for testing time-based requirements.
- Test accounts and seed data are created by scripts before execution.
- TLS certificates (self-signed is acceptable) are available for the deployment-like test.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment or database is unavailable for more than 4 hours.
- The build is so unstable that more than 30% of test cases are blocked.
- A critical defect in login or messaging blocks all further testing.

Resume testing when:

- Blocking defects are fixed and a new build passes the smoke test (register, login, send one message).
- The environment is restored and verified.

13. Test Case Management & Traceability

Test cases are listed in Appendix A and tracked in the repository with their status (Not run, Pass, Fail, Blocked). The RTM below links each SRS requirement to its test cases and extends the Test Case ID columns of SRS Appendix C. All 56 requirement items in scope have at least one test case; 37 test cases are defined.

Req. ID	Requirement (short)	Test case IDs
REQ-1	Register with unique username, email, password	TC-AUTH-01
REQ-2	Reject invalid or duplicate registration	TC-AUTH-02
REQ-3	Store passwords as salted bcrypt hashes	TC-AUTH-01

Req. ID	Requirement (short)	Test case IDs
REQ-4	Login and 24-hour JWT	TC-AUTH-03
REQ-5	Generic error and lock after 5 failures	TC-AUTH-04
REQ-6	WebSocket only with valid token	TC-AUTH-05
REQ-7	Logout	TC-AUTH-06
REQ-8	Open private conversation	TC-MSG-01
REQ-9	Real-time delivery over WebSocket	TC-MSG-02
REQ-10	Message length 1-2,000	TC-MSG-03
REQ-11	Store and deliver to offline user	TC-MSG-04
REQ-12	Display sender, time, status, own messages	TC-MSG-02
REQ-13	Sanitise content (XSS)	TC-MSG-05
REQ-14	Create group	TC-GRP-01
REQ-15	Group broadcast and storage	TC-GRP-02
REQ-16	Admin add/remove/rename, leave	TC-GRP-03
REQ-17	Sender name on group messages	TC-GRP-02
REQ-18	Only members read/send	TC-GRP-04
REQ-19	Online/offline within 2 s	TC-PRS-01
REQ-20	Typing indicator	TC-PRS-02
REQ-21	Heartbeat, offline within 40 s	TC-PRS-03
REQ-22	Auto reconnect with back-off	TC-PRS-04
REQ-23	Persist before ack	TC-HIS-01
REQ-24	Paged history (50)	TC-HIS-02
REQ-25	SENT/DELIVERED/READ status	TC-STS-01
REQ-26	Real-time status update	TC-STS-01
REQ-27	Unread count	TC-STS-02
REQ-28	In-app notification, tab title	TC-STS-02
REQ-29	Edit display name, status	TC-PRF-01
REQ-30	Change password	TC-PRF-02
REQ-31	Search users	TC-MSG-01
PERF-1	Latency 500 ms (LAN) / 1 s	TC-PERF-01
PERF-2	100 concurrent connections	TC-PERF-02
PERF-3	Login/registration 2 s	TC-PERF-02
PERF-4	Load 50 messages 1 s	TC-HIS-02
PERF-5	Initial page 3 s at 4 Mbps	TC-PERF-03
SAF-1	Write before ack	TC-HIS-01
SAF-2	Rate limit 20 per 10 s	TC-SEC-04
SAF-3	Fail gracefully	TC-PRS-04, TC-SAF-01
SAF-4	Backup and restore	TC-SAF-02
SEC-1	Authentication before use	TC-AUTH-03, TC-AUTH-05
SEC-2	bcrypt, no secrets in logs	TC-SEC-01
SEC-3	TLS 1.2+	TC-SEC-02
SEC-4	Input validation, injection prevention	TC-MSG-05
SEC-5	Authorisation per conversation	TC-GRP-04

Req. ID	Requirement (short)	Test case IDs
SEC-6	Origin check	TC-SEC-03
SEC-7	Token expiry, lockout	TC-AUTH-04
SEC-8	Privacy, OWASP Top 10	TC-SEC-05
BR-1	Only authenticated users chat	TC-AUTH-05
BR-2	Unique usernames/emails	TC-AUTH-02
BR-3	See only own conversations	TC-GRP-04
BR-4	Only admin manages group	TC-GRP-03
BR-5	1-2,000 chars, no editing	TC-MSG-03
BR-6	Lock after 5 failures	TC-AUTH-04
USABILITY	Register and send first message in 2 min (SRS 6.4)	TC-USB-01
RELIABILITY	No message loss, 4-hour run (SRS 6.4)	TC-REL-01

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed, passed, failed and blocked
- Requirement coverage (requirements with at least one passed test case)
- Defect count by severity and module, defect density, defect aging
- Unit test coverage of server logic (target 70%)
- Performance results: 95th percentile message latency, login time, history load time, connections sustained

Reports: execution status at the end of each sprint and the final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
Test Lead		
Team (Dev) Representative		
Project Guide		

Appendix A: Test Cases

Test cases cover the functional requirements and the performance, safety, security, usability and reliability requirements. Status of every case at the time of writing: Not run (the implementation is starting in sprint 1).

TC ID	Req. ID	Scenario and steps	Expected result	Type
TC-AUTH-01	REQ-1, REQ-3	Register with a valid username (3-20 chars), valid email and password "Test1234". Inspect the users collection.	Account created (201); passwordHash is a salted bcrypt hash ($2b$...), plain password not stored anywhere.	Functional, Integration
TC-AUTH-02	REQ-2, BR-2	Try registrations with: existing username, same username in different case ("Alice" vs "alice"), existing email, invalid email, password of 7 chars, password with no digit, password with no letter, username of 2 and 21 chars.	Every attempt rejected with a specific VALIDATION_ERROR or DUPLICATE_USER reason; no account created.	Functional, Negative
TC-AUTH-03	REQ-4, SEC-1	Log in once with username and once with email, both with the correct password. Decode the JWT.	Login succeeds in both cases; JWT is signed and its expiry is 24 hours after issue; WebSocket opens and Chat Home is shown.	Functional
TC-AUTH-04	REQ-5, SEC-7, BR-6	Enter a wrong password 5 times in a row, then a 6th time with the correct password; repeat after 15 minutes (use a shortened lock time in the test configuration).	Each failure shows only "Invalid credentials"; after the 5th the account is locked and the 6th attempt returns ACCOUNT_LOCKED; login works again after the lock expires.	Functional, Security
TC-AUTH-05	REQ-6, SEC-1, BR-1	Open a WebSocket (a) without sending auth, (b) with a tampered token, (c) with an expired token, (d) with a valid token. Call a REST endpoint without a token.	(a) closed after 5 s with code 4401, (b) and (c) AUTH_FAILED and closed, (d) auth:ok. REST call without token returns 401. Visitors cannot use chat functions.	Functional, Security
TC-AUTH-06	REQ-7	Click Logout while connected.	WebSocket closed, client token removed, Login screen shown, contacts see the user go offline; back button does not restore the session.	Functional
TC-MSG-01	REQ-8, REQ-31	Type "ali" in the search box (users Alice, ALICIA, Natalie exist); click a result to start a chat; repeat for the same user.	Partial, case-insensitive matches are listed; a private conversation is created the first time and the same one is reused afterwards.	Functional
TC-MSG-02	REQ-9, REQ-12	User A and user B are online. A sends "Hello". Observe both panes without refreshing.	Message appears in B's pane in real time with sender, server timestamp and status; A's own messages are visually distinct; order is chronological.	Functional, System
TC-MSG-03	REQ-10, BR-5	Send messages of 0 chars, spaces only, 1 char, 2,000 chars and 2,001 chars. Look for any edit option on a sent message.	1 and 2,000 chars accepted; empty, whitespace-only and 2,001 rejected with an error and not stored; no edit function exists.	Functional, Boundary
TC-MSG-04	REQ-11	B is offline. A sends three messages. B logs in later.	Messages are stored with status SENT; B receives them in order on connect; A's ticks change to DELIVERED.	Functional, Integration
TC-MSG-05	REQ-13, SEC-4	Send "<script>alert(1)</script>" and "<img src=x onerror=alert(1)>". Also log in with {"$ne": ""} as the password.	Both strings are displayed as plain text and no script runs, on the client and in stored data. The injection login is rejected with VALIDATION_ERROR or AUTH_INVALID_CREDENTIALS.	Security, Negative
TC-GRP-01	REQ-14	Create a group with valid name and 2 other members; try a 2-char name, a 51-char name, and only 1 other member.	Valid group created and creator is ADMIN; invalid attempts rejected with a clear reason; members get group:update.	Functional, Boundary
TC-GRP-02	REQ-15, REQ-17	Member posts in a group of 4 where 2 members are offline.	Online members get the message in real time with sender name; it is stored and delivered to offline members when they connect.	Functional, System
TC-GRP-03	REQ-16, BR-4	Admin adds, removes and renames; a non-admin tries the same; a member leaves the group and the group posts again.	Admin actions succeed and all members receive group:update; non-admin gets 403 FORBIDDEN_NOT_ADMIN; the member who left receives no further messages.	Functional, Security

TC ID	Req. ID	Scenario and steps	Expected result	Type
TC-GRP-04	REQ-18, SEC-5, BR-3	User C (not a member) reads a group/private conversation by ID via REST and sends to it via WebSocket.	Both rejected with FORBIDDEN_NOT_MEMBER; no content leaks; the conversation list shows only the user's own conversations.	Security, Negative
TC-PRS-01	REQ-19	A watches B's status while B logs in, then logs out. Measure the time to update.	Status changes to online/offline within 2 seconds each time; last-seen time shown for offline users.	Functional, Performance
TC-PRS-02	REQ-20	B types in the chat, then (a) stops for 3 s, (b) sends the message.	A sees "B is typing..."; indicator disappears after 3 s of inactivity and when the message is sent.	Functional
TC-PRS-03	REQ-21	Block the client's network without closing the browser (firewall rule). Observe the server and other clients.	Server detects the missed pong and marks the user offline within 40 seconds; contacts see the change.	Reliability
TC-PRS-04	REQ-22, SAF-3	Stop the server for 20 seconds while a client is open, then restart it. Send messages from another client meanwhile.	Client shows "Reconnecting..." and retries at 1 s, 2 s, 4 s ... up to 30 s; after reconnect it fetches missed messages with no duplicates and shows "Connected".	Reliability, System
TC-HIS-01	REQ-23, SAF-1	Send a message, wait for the SENT ack, then kill the server process (kill -9) and restart.	The acknowledged message is present in the database and in the history.	Reliability, Integration
TC-HIS-02	REQ-24, PERF-4	Seed 120 messages in a conversation. Open it, scroll up twice; time the first load.	Latest 50 shown; scrolling loads the next 50, then the remaining 20; first load takes 1 second or less.	Functional, Performance
TC-STS-01	REQ-25, REQ-26	A sends to B (online, conversation not open), then B opens the conversation.	A sees Sent, then Delivered when B's client receives it, then Read when B opens it; each change appears in real time.	Functional
TC-STS-02	REQ-27, REQ-28	A sends 3 messages to B while B has another conversation open.	B's unread badge shows 3, an in-app notification is shown and the tab title is updated; badge resets to 0 when B opens the conversation.	Functional
TC-PRF-01	REQ-29	Save display names of 40 and 41 chars and status messages of 100 and 101 chars.	40/100 accepted and visible to contacts; 41/101 rejected with VALIDATION_ERROR.	Functional, Boundary
TC-PRF-02	REQ-30	Change password with a wrong current password, with a weak new password, and with correct values; then log in with old and new password.	First two rejected; valid change confirmed; only the new password works afterwards.	Functional, Security
TC-PERF-01	PERF-1	Two clients on the local network exchange 200 messages (script); record send-to-display time. Repeat over a broadband connection.	At least 95% of messages take 500 ms or less on the LAN and 1 s or less on broadband.	Performance
TC-PERF-02	PERF-2, PERF-3	Run 100 simulated WebSocket clients (k6/Artillery) for 10 minutes, each sending a message every 5 s, while 20 users register and log in.	No message loss or server crash; login and registration complete within 2 seconds.	Performance, Load
TC-PERF-03	PERF-5	Load the initial page with network throttled to 4 Mbps (DevTools) on Chrome.	Page is usable within 3 seconds.	Performance
TC-SEC-01	SEC-2	Run registration, login, failed login and messaging flows; search server logs and error responses for passwords, hashes, tokens and message text. Check the bcrypt cost factor.	None of these appear in logs or responses; bcrypt cost factor is 10 or higher.	Security, Inspection
TC-SEC-02	SEC-3	Scan the hosted deployment with testssl.sh/openssl s_client; try http:// and ws:// against it.	TLS 1.2 or higher only; plain http/ws not served; wss:// connection works.	Security
TC-SEC-03	SEC-6	Open a WebSocket with a valid token but Origin: https://evil.example, using a script.	Handshake rejected; no presence or message events delivered.	Security, Negative
TC-SEC-04	SAF-2	Send 21 messages within 10 seconds from one user; also send a frame larger than 4 KB.	21st message rejected with RATE_LIMITED and the server stays responsive; oversized frame rejected or connection closed.	Security, Reliability
TC-SEC-05	SEC-8	Review the code, dependency list (npm audit) and network calls against the OWASP Top 10 checklist.	No data is sent to third parties; no high or critical vulnerabilities left open.	Security, Review

TC ID	Req. ID	Scenario and steps	Expected result	Type
TC-SAF-01	SAF-3	Stop MongoDB while a user sends a message and while another logs in; restart MongoDB.	Users see a clear "service unavailable" message; message is not acknowledged and no partial or corrupt record exists; service recovers after restart.	Reliability, Negative
TC-SAF-02	SAF-4	Take a database backup, delete a test conversation, restore from the documented procedure.	Data is restored exactly; procedure steps are accurate.	Operational
TC-USB-01	USABILITY	Five first-time users register and send a first message without help; repeat the main flows on a 360 px wide screen and in Chrome, Firefox and Edge.	At least 90% finish within 2 minutes; layout is usable at 360 px; behaviour is the same in all browsers.	Usability, Compatibility
TC-REL-01	RELIABILITY	Run the server for a 4-hour session with 10 active clients and some network drops.	No restart needed, no accepted message lost, clients recover automatically.	Reliability, Soak